Hey Buddy: A Multimodal Fitness Coach Providing Real-Time Form Feedback

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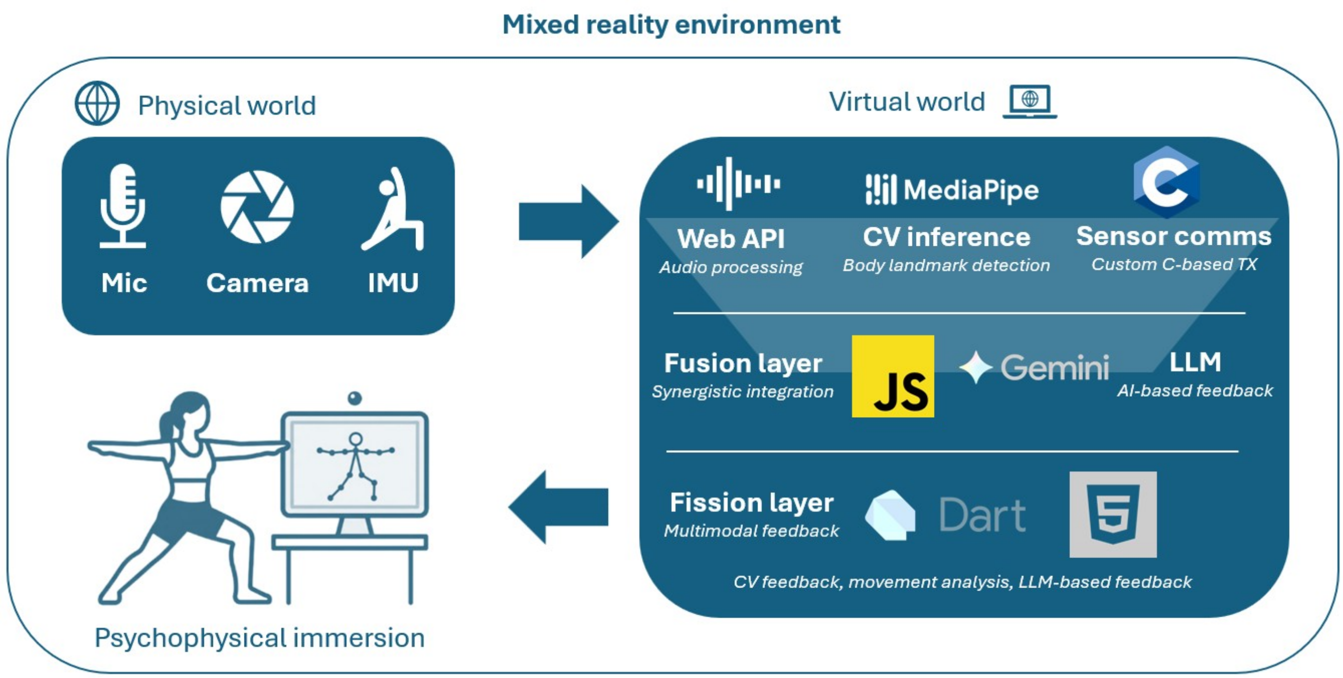

Figure 1: Overview of the Hey Buddy system architecture and multimodal loop. Sensor inputs (microphone, camera, IMU) are processed via speech recognition, pose detection, and LLM reasoning, and integrated in a fusion engine to generate real-time multimodal feedback (visual and audio) in a mixed-reality setup.

Abstract

Hey Buddy is a multimodal fitness coaching system that provides real-time feedback on exercise execution by integrating visual pose estimation, inertial sensing, and voice-based interaction. The system is designed to support users during physically demanding tasks by delivering context-aware feedback with minimal cognitive load.

This paper presents the system architecture and interaction design, and investigates whether real-time multimodal feedback improves squat execution under fatigue. We report results from a two-phase user study, including qualitative observations and a controlled within-subject experiment.

Our findings show that users achieved significantly greater squat depth when receiving feedback, with a large effect size, indicating that multimodal real-time coaching can meaningfully improve motor performance. We discuss design implications for real-time feedback systems and highlight directions for future multimodal fitness applications.

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Keywords

multimodal interaction, real-time feedback, fitness coaching, human-computer interaction, user evaluation

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1 Introduction

Correct execution of strength exercises is essential for both performance and injury prevention. However, most recreational athletes train without continuous expert supervision and must rely on mirrors, delayed video feedback, or subjective body awareness. These approaches are particularly limited under physical fatigue, where motor control degrades and the risk of improper form increases.

Recent advances in computer vision, speech interfaces, and large language models (LLMs) enable a new class of multimodal coaching systems capable of observing movement, interpreting user intent, and delivering adaptive feedback in real time. Despite this progress, many existing solutions focus on isolated modalities or static metrics, often resulting in fragmented, poorly timed, or cognitively demanding feedback.

In this work, we present *Hey Buddy*, a multimodal fitness coaching prototype designed to provide real-time, phase-aligned feedback during exercise execution. The system combines visual pose analysis, inertial sensing, and voice interaction, and uses a fusion-based architecture to generate context-aware coaching cues. Rather than presenting raw data, Hey Buddy translates multimodal input into actionable feedback aligned with the user's current movement phase.

The design is grounded in the CASE/CARE model of multimodal interaction, with a focus on synergistic modality integration to improve usability, robustness, and feedback quality.

To evaluate the system, we conducted a two-phase user study. A qualitative observation phase was used to identify usability issues, followed by a controlled within-subject experiment comparing feedback and no-feedback conditions under fatigue. We specifically investigate whether real-time multimodal feedback improves squat depth as a key indicator of exercise quality.

The results show a significant improvement in squat depth when feedback is provided, suggesting that multimodal coaching systems can have a measurable impact on physical performance.

This paper contributes (i) a multimodal coaching system integrating vision, audio, and inertial sensing, (ii) a real-time fusion architecture grounded in CASE/CARE principles, and (iii) empirical evidence demonstrating the effectiveness of multimodal feedback for improving exercise execution. **GitHub Repository:** https://github.com/SantiagoMartinezHernandez/HeyBuddy

2 Toolkit

As Digital Neuroscience students, we did not integrate a toolkit. Instead, we present a dedicated User Interaction section below.

3 Software Architecture

Hey Buddy is implemented as a modular, event-driven system that integrates computer vision, speech processing, inertial sensing, and large language models (LLMs) to enable real-time multimodal coaching during physical exercise.

Figure 2 illustrates the system pipeline, based on the *Multimodal Interaction Conceptual Framework* by Dumas et al. [1], adapted to the Hey Buddy system. The framework emphasizes real-time multimodal processing, semantic fusion, and coordinated feedback generation across heterogeneous modalities.

The core system is developed natively in Python, which handles pose estimation, kinematic computation, sensor processing, fusion logic, and feedback timing. Visual sensing is implemented using MediaPipe pose estimation, extracting joint landmarks and computing exercise-specific metrics such as hip depth, knee alignment, and back angle. Audio input is processed using OpenWakeWord for detecting the wake word "Hey Buddy", enabling hands-free interaction during exercise. Additionally, a custom web-based sensor implemented in C streams motion-related data to the backend, improving robustness and temporal resolution of movement tracking.

The system follows a pipeline architecture in which modality-specific modules operate independently and communicate via structured data streams. Each component produces intermediate representations (e.g., pose landmarks, detected events, or commands) that are passed to downstream modules through event-driven messaging. This enables low-latency processing while keeping components loosely coupled.

To provide a user-facing interface, the Python backend is integrated into a Flutter-based application, forming the Hey Buddy app, which runs on both desktop and Android devices. The Flutter frontend handles visualization, rendering of overlays, and user interaction, while the Python backend performs all computation.

Communication between frontend and backend follows a client–server model using a lightweight API layer. Messages are exchanged asynchronously, allowing continuous streaming of sensor data and real-time feedback delivery. This separation between computation and presentation ensures scalability, portability, and maintainability across platforms.

Visual feedback is rendered as overlays on the live camera feed, while auditory feedback is generated using text-to-speech. The architecture cleanly separates sensing, processing, communication, and presentation layers, supporting extensibility to additional exercises and modalities.

To ensure robust and low-latency exercise analysis, the system implements a structured motion processing pipeline. A bad repetition filter first removes noisy segments based on landmark visibility (threshold > 0.6) and temporal consistency. A repetition counter then detects complete squat cycles based on state transitions, while a short-term buffer maintains recent kinematic data to enable temporal smoothing and comparison across repetitions.

At the core of this pipeline is a finite-state machine modeling the squat as four phases (standing, descending, bottom hold, ascending). This allows the system to transform raw pose and sensor data into stable representations for real-time feedback and evaluation. Key joints include shoulders (11,12), hips (23,24), knees (25,26), and ankles (27,28) as defined by the MediaPipe pose model.

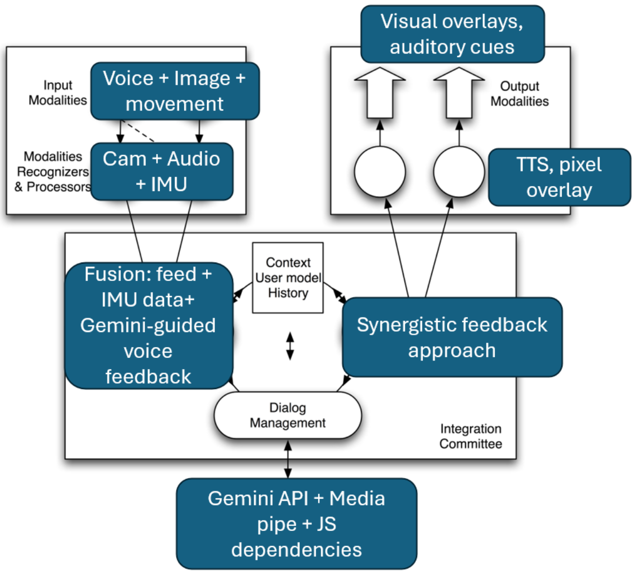

Figure 2: Fusion-based multimodal architecture of Hey Buddy, adapted from the Multimodal Interaction Conceptual Framework by Dumas et al. [1]. The system integrates voice, vision, and motion into a centralized fusion engine that delivers real-time visual and auditory feedback.

4 Fusion Engine

Multimodal fusion in Hey Buddy is based on the CASE/CARE model introduced by Nigay and Coutaz [2] and further elaborated by Dumas et al. [1], which explains both how and why modalities are combined.

From a **CASE perspective** (how modalities are combined), the system performs semantic-level fusion through a centralized fusion engine. Visual pose data (MediaPipe via OpenCV), inertial motion data from a custom C-based sensor system (ESP32), and verbal input (wake-word-triggered interaction) are jointly integrated into a unified representation of the user's exercise state. Data streams are processed in parallel using JavaScript and Dart, enabling real-time synchronization and multimodal query resolution.

From a **CARE perspective** (why modalities are combined), each modality contributes complementary strengths: vision provides precise spatial and kinematic information, inertial sensing improves temporal consistency and robustness (e.g., sacrum orientation and acceleration), and audio enables hands-free interaction. Additionally, large language models (LLMs) enhance interpretability by generating expressive, context-aware feedback. This combination increases reliability, usability, and overall system robustness compared to unimodal approaches.

Hey Buddy primarily operates in the *synergistic* fusion quadrant, where combined modalities produce insights that are not achievable through any single modality alone. For example, knee misalignment can be detected visually, motion consistency validated via sensor input, and corrective instructions generated through audio feedback aligned with the current movement phase.

The fusion engine integrates modality-specific outputs together with dialog context and short-term history to infer the current system state. A dialog manager interfaces this representation with the Gemini LLM (models/gemini-2.0-flash via Google API). In addition to predefined heuristic-based feedback, the system includes an LLM fallback mechanism that processes user queries outside programmed routines. In such cases, multimodal data is packaged into a structured JSON representation and passed to the LLM for higher-level analysis.

The resulting feedback is passed to the fission layer, implemented using Dart and HTML, where it is synchronized across modalities and delivered as real-time audio and visual feedback, along with near-real-time LLM-generated responses.

The squat movement is explicitly modeled using a finite-state machine consisting of four discrete states: standing, descending, bottom hold, and ascending. Transitions between these states are determined using joint kinematics such as hip displacement, joint angles, and movement velocity. For example, a transition from standing to descending is triggered when downward hip motion exceeds a predefined threshold, while the bottom hold state is detected when vertical movement stabilizes near the lowest hip position.

This explicit state representation enables temporally aligned feedback generation and supports consistent repetition counting and quality assessment. Exercise quality is evaluated using lightweight heuristics with explicit thresholds: squat depth is estimated from relative hip–knee positions, knee valgus is detected when inward knee displacement exceeds approximately 5% relative to hip and ankle alignment, and excessive torso lean is identified using phase-dependent thresholds (approximately 55% during descent and 65% during ascent). Inertial sensing complements visual tracking by providing sacrum orientation, where forward pitch beyond roughly 10–15 degrees indicates loss of neutral posture, and wrist-based acceleration estimates support analysis of movement speed during ascent.

5 User Interaction

User interaction in Hey Buddy is designed around real-time, low-attention feedback that supports users during physically demanding tasks without interrupting movement. Interaction is multimodal, combining speech-based commands with continuous visual and auditory feedback.

A key interaction pattern is **on-demand posture feedback**. Users can query the system (e.g., "How's my posture?"), prompting the system to analyze the current movement state and provide detailed corrective feedback. For example, the system may indicate insufficient squat depth or detect inward knee collapse, followed by actionable instructions such as pushing knees outward or adjusting torso position. Feedback is directly tied to the current movement phase, ensuring temporal relevance.

The system also supports **progress and quality feedback**, where users can ask about improvements over time (e.g., "Am I doing better than before?"). The system leverages short-term memory to compare current performance with previous repetitions, providing feedback such as increased stability or improved alignment. This enables reflective training and reinforces learning.

Another interaction mode is **time-based feedback**. Users can request temporal information (e.g., "How long have I been holding this squat?"), and the system provides real-time metrics such as duration or countdown-based suggestions. This gives users awareness of performance over time and supports decision-making (e.g., whether to continue or stop).

But the best example of synergistic interaction is the isometric hold position: when the application detects that the user is in a squat position, and the user uses the required voice instruction, the it will trigger the iso-chronometer, that shows the time the user spends in a squat position.

Finally, the system enables **motivational and task-oriented feedback** on demand, such as remaining repetitions or encouragement based on performance. This maintains engagement without requiring constant attention.

Visual feedback is continuously displayed via overlays, including pose skeletons, joint angles, and highlighted error regions (e.g., misaligned knees). Auditory feedback complements this by delivering concise instructions through text-to-speech, allowing users to maintain focus on physical execution rather than the interface.

Internally, feedback generation is organized into distinct intent types to ensure consistent interaction. Direct feedback intents provide immediate corrective cues aligned with the current movement phase, time-based intents support countdowns and duration tracking (e.g., during bottom holds), and motivational intents provide reinforcement during repeated execution. A fallback mechanism using the Gemini LLM handles more complex or ambiguous queries.

To ensure reliable interaction, the system includes safeguards such as wake-word activation ("Hey Buddy"), intent validation, and lightweight filtering to prevent unintended or noisy inputs.

Overall, the interaction design emphasizes context-aware, phase-aligned feedback, short-term memory for comparison, and user-driven queries. This results in a multimodal coaching experience that is both informative and minimally intrusive, supporting effective training under real-world conditions.

6 Applications

Hey Buddy is designed as a multimodal fitness assistant for exercises that require precise posture control under physical load. In this prototype, the system focuses on squat execution from a lateral view, monitoring hip depth, knee alignment, and back posture.

Prior work has shown that augmented feedback can significantly enhance training performance. Weakley et al. [3] demonstrated in a systematic review and meta-analysis that high-frequency feedback can increase acute resistance training performance by up to 8.4%. Validating these findings, Hey Buddy targets home-training environments where human coaches are typically absent, delivering continuous, real-time multimodal feedback to support proper execution.

During exercise, users receive immediate corrective cues such as warnings for insufficient squat depth or inward knee collapse, enabling them to adjust movement in real time rather than relying on delayed feedback. This aligns with evidence suggesting that temporally precise feedback is critical for effective motor learning.

Beyond squats, the architecture generalizes to other compound movements (e.g. deadlifts or planks) by adapting movement-specific kinematic rules. The system is intended for home training scenarios, where it provides scalable, coach-like guidance that can both improve performance and reduce injury risk without requiring continuous expert supervision.

7 User Evaluation

The user evaluation followed a two-phase approach aligned with a user-centred design methodology. The goal was to both identify early usability issues and quantitatively assess the effect of real-time multimodal feedback on exercise form quality.

7.1 Phase 1: Qualitative Observation

The first phase consisted of direct observation sessions with 5 participants. Participants performed squat exercises using Hey Buddy while the evaluator observed silently. A retrospective think-aloud protocol was used between sets to collect user impressions without interfering with motor execution.

Observed events were logged using a structured coding sheet capturing interaction events, system errors, and multimodal fusion failures. Insights from this phase informed refinements to feedback timing, feedback clarity, and stability improvements for the controlled experiment.

7.2 Phase 2: Controlled Experiment

The second phase aimed at verifying the following hypothesis: « Participants using the real-time visual feedback display (moving hip-height marker against a personal ROM reference line) will achieve a significantly greater mean squat depth, expressed as % of their personal ROM baseline, compared to the no-feedback condition. ». The protocol relied on a within-subject experimental design with 8 participants. Each participant completed squat sessions under two conditions: *feedback ON* (FB) and *feedback OFF* (NFB).

Fatigue was operationalized using a standardized protocol consisting of two sets of ten squats at 70% of one-repetition maximum (1RM), increasing form degradation pressure compared to the initial prototype.

The primary dependent variable was squat depth, measured as normalized range-of-motion (ROM / height %) and converted to absolute depth in cm. Conditions were counterbalanced across participants to control learning effects and fatigue carryover.

7.3 Analysis and Results

A paired-samples t-test was conducted to compare squat depth between feedback and no-feedback conditions. Results indicate a statistically significant improvement in squat depth when feedback was provided.

The analysis revealed a significant effect of feedback on squat depth ($t(7) = 9.635$, $p < 0.001$), with a large effect size (Cohen's $d = 3.407$). On average, participants reached greater depth both in relative (ROM) and absolute (cm) terms when using Hey Buddy.

These findings support the hypothesis that real-time multimodal feedback improves exercise execution under fatigue conditions.

7.4 Discussion of Findings

The qualitative observations from Phase 1 highlighted several usability issues, including timing inconsistencies of feedback and

Table 1: Paired-samples t-test results comparing feedback (FB) and no-feedback (NFB) conditions.

Metric	Value
Number of participants (N)	8
Mean FB ROM/Height (%)	36.897
Mean NFB ROM/Height (%)	35.422
Mean difference (%)	1.475
Mean difference (cm)	2.522
SD of differences	0.433
95% Confidence interval	[1.175, 1.775]
t-statistic	9.635
Degrees of freedom	7
p-value	< 0.001
Cohen's d	3.407
Effect size	Large

occasional fusion errors. These were partially addressed before Phase 2 and likely contributed to the strong observed effect.

The quantitative results demonstrate that even a lightweight multimodal feedback system can produce measurable improvements in motor execution. However, the controlled setting and relatively small sample size limit generalizability.

Future work should validate these findings in more ecologically valid training scenarios and expand evaluation metrics beyond squat depth to include alignment and stability measures.

8 Conclusion

We presented *Hey Buddy*, a multimodal fitness coaching system that combines visual pose estimation, inertial sensing, and voice interaction to provide real-time feedback during exercise. The system leverages a fusion-based architecture to deliver context-aware, low-latency feedback aligned with the user's movement.

Through a two-phase user evaluation, we demonstrated that real-time multimodal feedback significantly improves squat performance under fatigue. Participants achieved greater squat depth in the feedback condition, with a large effect size, supporting the effectiveness of multimodal coaching for enhancing motor execution.

These findings highlight the potential of multimodal interaction systems to move beyond passive tracking toward active performance guidance in physically demanding tasks.

Despite these promising results, several limitations must be addressed. First, audio interaction showed reduced robustness, as users experienced difficulties with wake-word activation and command recognition at greater distances. This suggests the need for improved microphone placement, such as wearable or sensor-integrated solutions. Second, the absence of a dedicated preprocessing pipeline led to jitter in landmark estimation under varying environmental conditions; future systems should incorporate stabilization techniques while carefully balancing latency constraints. Third, the vision-based pose estimation degraded when body landmarks were occluded, indicating that tighter fusion with IMU data or multi-camera setups could improve robustness. Finally, device-dependent usability issues, particularly on smartphones, revealed challenges with interaction design and tracking accuracy; adaptive interfaces and multimodal confirmation mechanisms may help mitigate these effects.

Future work should therefore extend the evaluation to more diverse user groups and longer-term training scenarios, while incorporating additional performance metrics such as alignment and movement stability. In the long term, we envision multimodal coaching systems like *Hey Buddy* as scalable, personalized training assistants that provide expert-level feedback in everyday environments without requiring human supervision.

Author Contributions

The project was developed collaboratively, with each team member contributing to different components of the multimodal system.

Andres Martinez Hernandez and Simon Krummenacher were responsible for the audio and voice interaction pipeline, including speech processing, integration of the Gemini API, and dialog-based feedback generation.

Mathilde Voyame and Sebastien Pierret focused on the visual processing pipeline, including video acquisition and pose estimation using MediaPipe.

Andres Martinez Hernandez additionally contributed to sensor integration and the deployment of the system within a Flutter-based application.

All authors contributed to the system design, integration, experimentation, and writing of the final report.

AI Usage Statement

Parts of this work were supported by the use of AI-based tools (e.g., Microsoft Copilot) for language refinement, structuring, and code debugging.

All conceptual design, implementation, and evaluation were carried out by the authors. The authors take full responsibility for the content.

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A Online Resources

The source code and technical documentation for the Hey Buddy prototype are publicly available to support transparency and reproducibility.

We additionally provide a demonstration video showing the Hey Buddy system in use, highlighting real-time multimodal feedback and overall interaction flow during a workout.